

Swathi Nachiar Manivannan

PhD candidate in Ecology and Evolutionary Biology, Yale University
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I am an evolutionary biologist interested in understanding the underlying mechanisms that shape the capacity of microbes to adaptively evolve.

EDUCATION

Ph.D. in Ecology and Evolutionary Biology Yale University

2022—*present*
New Haven, CT, USA

Proposed thesis: *The path(s) to optimality: towards a mechanistic understanding of evolvability in microbes*

Thesis Committee: C. Brandon Ogbunugafor (advisor), Richard O. Prum, Günter P. Wagner, Jeremy Draghi

M.Sc. in Ecology and Evolutionary Biology Yale University

2022—2024
New Haven, CT, USA

B.A. (Hons) in Natural Sciences (Genetics) Homerton College, University of Cambridge

2018—2021
Cambridge, UK

Thesis: *The genetics of hybridisation: What systems biology-based models can tell us* (Advised by John J. Welch)

RESEARCH EXPERIENCE

Graduate Research Fellow (Advisor: C. Brandon Ogbunugafor) Department of Ecology & Evolutionary Biology, Yale University

2022—*present*

Research Officer

Bioinformatics Institute, A*STAR, Singapore (Advisor: Sebastian Maurer-Stroh)

2022

Genome Institute of Singapore, A*STAR, Singapore (Advisor: Swaine Chen)

2021—2022

Undergraduate Researcher

Part II Genetics Student, University of Cambridge (Advisor: John J. Welch)

2020—2021

International Student Internship Program, Princeton University (Advisor: Simon A. Levin)

2020

Genome Institute of Singapore, A*STAR, Singapore (Advisor: Wei Leong Chew)

2019

HONOURS AND AWARDS

Associates in Teaching, Yale University

2026

CRC Mellon Pedagogy Fellowship, RITM, Yale University

2025

Sterling Prize Fellowship, Yale University

2022—2024

A*STAR National Science Scholarship (PhD)

2022—*present*

David Thompson Scholarship, Homerton College, University of Cambridge

2019

A*STAR National Science Scholarship (BS)

2018—2021

GRANTS

Yale Institute for Biospheric Studies (YIBS) Small Early Grant (\$3000)

2023

PUBLICATIONS

Peer-reviewed articles

6. **Manivannan, S.N.**, & Ogbunugafor, C.B. (2026). *Deconstructing empirical fitness seascapes across scales of granularity*. In press at G3: Genes, Genomes, Genetics.
5. Surasinghe, S., **Manivannan, S.N.**, Scarpino, S.V., Crawford, L., Ogbunugafor, C.B. (2026). *Structural causal influence (SCI) captures the forces of social inequality in models of disease dynamics*. In press at Biology Letters [[pre-print](#)].
4. **Manivannan, S.N.**, & Ogbunugafor, C.B. (2026). *Maynard Smith's analogy, realized: Common ancestry constrains evolutionary percolation through protein space*. PNAS, 123(21), e2610113123.
3. **Manivannan, S.N.**, Diaz Arenas, C., Grubaugh, N.D. & Ogbunugafor, C.B. (2025). *The importance of epistasis in the evolution of viral pathogens*. Virus Evolution, 11(1), veaf091.
2. **Manivannan, S.N.** (Advised by Levin, S.A.) (2023). *Modelling the evolutionary dynamics of an infectious disease with an initial asymptomatic stage with*

recovery. SIAM Undergraduate Research Online, 16.

- Chen, L., Park, J.E., Paa, P., Rajakumar, P.D., Prekop, H-T., Chew, Y.T., **Manivannan, S.N.**, & Chew, W.L. (2021). *Programmable C:G to G:C genome editing with CRISPR-Cas9-directed base excision repair proteins*. Nature Communications, 12(1).

Pre-prints

- Surasinghe, S.*, **Manivannan, S.N.***, Crawford, L. & Ogbunugafor, C.B. *Classification and regression trees clarify the role of epistasis and environment in genotype–phenotype maps*.

Magazine article(s)

- Manivannan, S.N.** *The construction of a metropolis, or the cost of uprooting*. The Yale Environmentalist Spring 2023 (p. 66-67).
- Giovanetti-Singh, G.*, Kent, R.*, **Manivannan, S.N.*** October 2021. *Hidden Figures*. BlueSci Issue 52 (p.18-23).

*denotes shared first author

PRESENTATIONS

Talks

Ecology & Evolutionary Biology Graduate Student Symposium, Yale University	2026
† Ecology and Evolution of Infectious Diseases Early Career Researchers Symposium, Yale University	2026
† Graduate Research in Ecology, Behaviour and Evolutionary Biology, Rutgers University	2026
** Fisher's Fishes	2025
Ecology & Evolutionary Biology Graduate Student Symposium, Yale University	2024
** Li-Richter lab group, Institute of Ecology & Evolution, Universität Bern	2024

Posters

Population, Evolutionary, and Quantitative Genetics (PEQG) Conference	2026
QBio/PEB Symposium, Yale University	2026
Molecular Mechanisms in Evolution, Gordon Research Conference & Seminar	2025

† lightning talk, ** invited talk

TEACHING EXPERIENCE

Yale University

<i>(forthcoming)</i> Instructor on record <i>Biology of Humans (MCDB/HSHM 4690)</i> , Yale University	2027
Guest Lecturer <i>Evolutionary Biology (EEB 2225)</i> , Yale University	2026
Certificate of College Teaching Preparation, Yale University	2025
McDougal Graduate Teaching Fellow Poorvu Center for Teaching and Learning, Yale University	2024—2026
Teaching Fellow	
<i>Evolution and Medicine (E&EB 335)</i> , Yale University	2023
<i>Principles of Ecology and Evolutionary Biology (BIOL 104)</i> , Yale University	2023
<i>Biology of Terrestrial Arthropods (E&EB250, E&EB251L)</i> , Yale University	2022

University of Cambridge

Volunteer Teaching Assistant <i>A Level Chemistry</i> Long Road Sixth Form College (STIMULUS Cambridge)	2019—2020
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PROFESSIONAL DEVELOPMENT

SLiM Workshop, American Natural History Museum	2024
Complexity Global School for Emerging Political Economics, Santa Fe Institute and Universidad de los Andes	2024
Evolutionary Biology Graduate Student Workshop, Department of Biology, University of Virginia	2023

MENTORSHIP

Undergraduate Senior Thesis

Kemper Lowry (Yale Anthropology and E&EB'25)

Women in Science at Yale (WISAY)

1 undergraduate, 2 Master's students

OUTREACH AND COMMUNICATION

Contributing Writer

The Yale Environmentalist, BlueSci, Varsity (Features), The Cambridge Language Collective

Outreach

Co-facilitator, Cold Spring School visit to the Peabody Museum, Yale Initiative for Science and Society 2026

Talk Coordinator for Science in the News, Yale Science Communication (*A graduate student organisation*) 2023

PROFESSIONAL SERVICE

Department Service

Graduate Student Jobs coordinator, E&EB Department, Yale University 2025—*present*

Graduate Student Symposium coordinator, E&EB Department, Yale University 2026

Presentation Judge, Undergraduate Student Symposium, E&EB Department, Yale University 2025

Hutchinson Speaker Seminars coordinator, E&EB Department, Yale University 2023—2024, 2026

Institutional Service

Co-facilitator, Course (Re)Design Institute, Poorvu Center for Teaching and Learning, Yale University 2026

Organising Team, Spring Teaching Forum, Poorvu Center for Teaching and Learning, Yale University 2025, 2026

International Committee Chair and Student Representative (E&EB), Graduate Student Assembly, Yale University 2024—2025

International Students' Representative, Women's Campaign, Cambridge Students' Union 2020—2021

International Officer, Homerton Union of Students, Homerton College, University of Cambridge 2019—2020

Peer Review

Nature Communications (1)

Other Service

Tamil Translator, Covid-19 Migrant Support Coalition (CMSC) in Singapore 2020—2021